

Modern Data Engineering for AI Systems

Capstone Rubric & Submission Requirements

SDAIA Academy, delivered via Learning Space | 5-day capstone | Trainer: Mohammed Albeladi

1. Capstone Rubric

100 points total | Pass mark: 60 or above. The capstone integrates all five days into one working pipeline.

#	Deliverable	Pts	What is required
1	Ingestion	20	Kafka producer and consumer with schema validation at the ingestion boundary (e.g. a Pydantic data contract). Malformed records routed to a quarantine zone or dead-letter topic with the rejection reason recorded.
2	Delta Lakehouse	25	Bronze / Silver / Gold layers using Delta Lake, with a real MERGE (upsert) keyed on a business key, and demonstrated schema enforcement . Gold must be a genuine aggregate, not a copy of Silver.
3	RAG Pipeline	25	Document chunking, embeddings, a vector store, hybrid search (dense vector + keyword/BM25, fused e.g. with Reciprocal Rank Fusion), and reranking (cross-encoder). Answers grounded in retrieved context with citations.
4	Orchestration	15	An Airflow DAG wiring every stage together with correct task dependencies, so a failed quality gate halts the pipeline before downstream stages run.
5	Quality Gate + Lineage	15	Great Expectations checks that actually gate the pipeline, plus OpenLineage START / COMPLETE / FAIL events emitted per stage.

How it is evaluated

- **Use the real library for each deliverable.** Credit is given for kafka-python / confluent-kafka, pyspark + delta-spark (or deltalake), a real vector store, apache-airflow, great-expectations, and openlineage-python.
- **A simulation does not earn the deliverable**, however well written - a queue standing in for Kafka, a custom orchestrator class instead of Airflow, or pandas/Parquet instead of Delta will not be credited.
- **Run your code and keep the output.** Executed notebooks with captured output, or logs of a real run, are the evidence that each stage actually works - not just that the code exists.
- **Prove the failure paths, not just the happy path.** Show malformed records actually being rejected, and a bad write actually being refused by schema enforcement.

2. GitHub & Documentation Requirements

These apply to **every project**, in addition to the rubric above. **They are part of how projects are evaluated.**

2.1 Mandatory

- Every trainee must **create and activate a GitHub account** if they do not already have one.
- All AI-related training projects must be **uploaded to GitHub**, kept documented and continuously updated.
- A project not published to GitHub as described here is **not a complete submission**.

2.2 Every project repository must include

Requirement	What it means in practice
Clear, comprehensive project description	What the project does, the problem it solves, and its scope - visible from the repo landing page.

Requirement	What it means in practice
Professional README	Explains the project idea and how to run and use it : prerequisites, install/setup steps, how to execute, and expected output.
Proper technical documentation	Architecture or pipeline overview, key components/modules, and any configuration or environment variables required.
Good Git version-control practices	Meaningful commit messages, incremental commit history (not a single bulk upload), sensible repo structure, and a .gitignore that excludes secrets and generated files.
Training program attribution	State clearly which training program the project was completed under (program name, and the cohort/session dates).
Link to SDAIA Academy on GitHub	Reference https://github.com/SDAIAAcademy in the README where relevant.

2.3 Encouraged: supporting the Saudi tech community

Trainees are encouraged to support outstanding Saudi projects on GitHub by:

- **Starring** high-quality repositories.
- **Following** Saudi accounts and repositories.
- **Contributing** to open-source projects.
- Engaging through **Fork**, **Pull Requests**, and **Issues** where appropriate.
- **Sharing** standout projects with the wider tech community.

Note: community engagement is encouraged and looked on favourably, but it is not scored against the 100-point rubric.